Mathematics Stage 3 -- Year B -- Unit 35 Addition & Subtraction Strategies, Decimals & Percentages

KLA: Mathematics | Stage 3 | Year 6 | Term: ___________ | Duration: 2-3 weeks | Teacher: _______________ | Year: _______

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-AR-01	Selects and applies appropriate strategies to solve addition and subtraction problems.
MA3-RN-03	Determines percentages of quantities, and finds equivalent fractions and decimals for benchmark percentage values.
MA3-NSM-02	Measures and compares duration, using 12- and 24-hour time and am and pm notation.
MA3-MR-02	Constructs and completes number sentences, applying the order of operations to calculations.
MA4-FRC-C-01	Stage 4 (group A): represents and operates with percentages — percentage increase/decrease, financial applications and the unitary method.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Number & time recall (single procedures) Find 10% (÷10), 25% (÷4) and 50% (half) of a quantity. Express thousandths as decimals; partition decimals by place value. Add and subtract decimals up to 3 decimal places. Use start and finish times to calculate elapsed time.	**Strategies, percentages & problem solving (applying)** Use place value to add/subtract 3+ numbers; decide when a calculator is efficient. Apply efficient strategies (levelling, addition for subtraction, constant difference, bridging). Solve decimal word problems and justify the chosen strategy. Represent percentages of quantities as fractions/decimals; find 10/25/50% discounts. Round to estimate and check the reasonableness of answers. Solve multi-step word problems requiring more than one operation.	**Extended -- Stage 4 Percentages** Find any percentage of a quantity, not just 10/25/50% (MA4-FRC-C-01). Calculate percentage increases and decreases, including discounts and 10% GST (MA4-FRC-C-01). Find the whole (original amount) from a given percentage using the unitary method (MA4-FRC-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students compare, evaluate and communicate the addition and subtraction strategies they choose, using clear mathematical language.
Understanding and fluency	Students add and subtract large numbers and decimals accurately, and find percentages of quantities.
Reasoning	Students justify why a strategy is appropriate and use estimation to judge the reasonableness of solutions.
Problem solving	Students solve multi-step word problems involving money, time and percentages, and (Stage 4) percentage increase/decrease problems.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band placements / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 35 | NESA Mathematics K-10 Syllabus (2022)

Unit 35 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Code	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Find 10%, 25% and 50% of a quantity.	RN-B percentages: 10% is one-tenth of 100%; equate 10% to ÷10, 25% to ÷4, 50% to half MA3-RN-03 Year B	**Mini Lesson 6**		
R2	Express thousandths as decimals; partition decimals by place value.	RN-A decimals: express thousandths as decimals; use place value to partition decimals MA3-RN-03 Year A	**Mini Lesson 4**		
R3	Add and subtract decimals up to 3 decimal places.	AR-B: model the addition and subtraction of decimals up to 3 decimal places MA3-AR-01 Year B	**Mini Lesson 5**		
R4	Use start and finish times to calculate elapsed time.	NSM-B time: use start and finish times to calculate the elapsed time of events MA3-NSM-02 Year B	**Mini Lesson 2**		
Y1	Add/subtract 3+ numbers with place value; decide when a calculator is efficient.	AR-A: add/subtract 3+ numbers with different digit lengths; determine when a calculator is efficient; compare strategies MA3-AR-01 Year A	**Mini Lesson 1**		
Y2	Apply efficient strategies (levelling, constant difference, bridging).	AR-A: apply known strategies such as levelling, addition for subtraction, constant difference and bridging MA3-AR-01 Year A	**Mini Lesson 3**		
Y3	Solve decimal word problems and justify the strategy.	AR-B: solve word problems involving add/subtract of decimals to 3 dp; justify the strategy used MA3-AR-01 Year B	**Mini Lesson 9**		
Y4	Represent percentages of quantities as fractions/decimals; find 10/25/50% discounts.	RN-B percentages: represent common percentages of quantities and lengths as fractions and decimals; estimate discounts of 10/25/50% MA3-RN-03 Year B	**Mini Lesson 7**		
Y5	Round to estimate and check the reasonableness of answers.	AR-A: round to obtain estimates; use estimation to check the reasonableness of solutions MA3-AR-01 Year A	**Mini Lesson 8**		
Y6	Solve multi-step word problems requiring more than one operation.	AR-B / MR-B: solve multistep word problems, including problems that require more than one operation MA3-AR-01 / MA3-MR-02 Year B	**Mini Lesson 10**		
G1	Find any percentage of a quantity.	Stage 4 FRC-A: apply knowledge of percentages to calculate quantities in various contexts MA4-FRC-C-01 Stage 4 (group A)	**Mini Lesson 11**		
G2	Calculate percentage increase and decrease, incl. discounts and GST.	Stage 4 FRC-A: calculate percentage increases/decreases; financial applications incl. GST, profit and loss MA4-FRC-C-01 Stage 4 (group A)	**Mini Lesson 12**		

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G3	Find the whole from a percentage (reverse percentage, unitary method).	Stage 4 FRC-A: solve real-life percentage problems using the unitary method MA4-FRC-C-01 Stage 4 (group A)	**Mini Lesson 13**		

 AR = Additive relations. RN = Represents numbers (decimals/percentages). MR = Multiplicative relations. NSM = Non-spatial measure (time). FRC = Fractions, decimals & percentages (Stage 4). A/B = syllabus content groups (B is the new add/subtract content for this Year-B unit).

Unit 35 -- Mini Lesson Sequence

Addition & Subtraction Strategies, Decimals & Percentages | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 2-6	Yellow band -- On-level -- Mini Lessons 1-10	Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 35. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Adding and subtracting large numbers

Y1 | Yellow band | DoE Lessons 1 & 2 (adapted)

Duration: 30 min | **Outcomes: Y1** | Syllabus: MA3-AR-01, MAO-WM-01 | AR-A: add/subtract 3+ numbers; choose efficient tools

Linked to: DoE Lessons 1 & 2 (adapted) -- Lessons 1 & 2 -- Place value strategies for addition and subtraction | Key resources: Resource 1 - shopping receipt, place-value charts, calculators

Learning intention	Success criteria
Students are learning to: use place value to add or subtract 3 or more numbers and decide when a calculator is efficient	Students can: * I can line up numbers by place value to add or subtract them. * I can add 3 or more numbers with different numbers of digits. * I can decide when a calculator is the most efficient tool.

Mini masterclass structure (30 min)

Activate (4 min): Show a shopping receipt and ask students to total three items mentally, then check with place value.

Model (10 min): Add 1240 + 56 + 308 by lining up place values; subtract 2050 - 875. Discuss when a calculator would be quicker (many large numbers).

Guided practice (12 min): Students add and subtract sets of 3+ numbers, choosing mental, written or calculator methods and explaining why.

Check (4 min): Exit question: 3406 + 78 + 920 = ? Would you use a calculator? Why?

Key vocabulary	Differentiation
place value, column, regroup, efficient, strategy, calculator, estimate	**Support:** Provide a place-value grid so digits line up; start with two numbers. **Extension:** Add a column of five 4-digit numbers and check with an estimate.

Assessment	Teaching notes (annotate here)
Students add and subtract several numbers efficiently — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Calculating elapsed time [HANDS-ON]

R4 | Red band | DoE Lessons 1, 7 & 8 (adapted)

Duration: 25-30 min	**Outcomes: R4**	Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-B: use start and finish times to find elapsed time

Linked to: DoE Lessons 1, 7 & 8 (adapted) -- Lessons 1, 7 & 8 -- Duration and timetables | Key resources: Geared clock face, time number line, timetable cards

Physical materials needed: geared clock face, time number line, timetable cards

Learning intention	Success criteria
Students are learning to: use start and finish times to calculate the elapsed time of an event	Students can: * I can find how long an event lasts from its start and finish times. * I can bridge through the hour to count the minutes. * I can use a time number line to keep track.

Mini masterclass structure (25-30 min)

Activate (4 min): A film starts at 3:45 pm and finishes at 5:20 pm. Ask how long it lasted.

Model (10 min): Use a time number line: 3:45 -> 4:00 (15 min) -> 5:00 (1 h) -> 5:20 (20 min) = 1 h 35 min. Bridge through the hour.

Guided practice (10 min): Students find elapsed times for events and timetable entries using a number line.

Check (4 min): Exit question: a bus leaves at 9:50 am and arrives at 11:15 am. How long is the trip?

Key vocabulary	Differentiation
elapsed time, duration, start time, finish time, bridge, am, pm, number line	**Support:** Use a geared clock and count on in 5-minute jumps to the next hour first. **Extension:** Solve elapsed-time problems that cross 12-hour boundaries and use 24-hour time.

Assessment	Teaching notes (annotate here)
Students calculate elapsed time — links to R4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Efficient add/subtract strategies

Y2 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30 min	**Outcomes: Y2**	Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: levelling, addition for subtraction, constant difference, bridging

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Mental and written strategies | Key resources: Number lines, mini whiteboards

Learning intention	Success criteria
Students are learning to: apply known strategies such as levelling, constant difference and bridging, and choose an efficient one	Students can: * I can use bridging to add through a ten or hundred. * I can use constant difference to make a subtraction easier. * I can pick an efficient strategy and explain why.

Mini masterclass structure (30 min)

Activate (4 min): Show 47 + 29. Ask how rounding 29 to 30 (then adjusting) makes it easier.

Model (10 min): Bridging: 47 + 29 = 47 + 30 - 1 = 76. Constant difference: 83 - 48 = 85 - 50 = 35. Levelling for addition.

Guided practice (12 min): Students solve additions and subtractions with different strategies and decide which is most efficient.

Check (4 min): Exit question: use constant difference to work out 72 - 39.

Key vocabulary	Differentiation
strategy, bridging, levelling, constant difference, compensation, efficient, mental	**Support:** Model each strategy on a number line; start with adding through a ten. **Extension:** Choose the most efficient strategy for a set of mixed problems and justify each choice.

Assessment	Teaching notes (annotate here)
Students apply efficient add/subtract strategies — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Thousandths and partitioning decimals

R2 | Red band | DoE Lesson 3 (adapted)

Duration: 25 min	**Outcomes: R2**	Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: express thousandths as decimals; partition decimals

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Decimal place value | Key resources: Place-value chart, decimal cards

Learning intention	Success criteria
Students are learning to: express thousandths as decimals and partition a decimal by place value	Students can: * I can write thousandths as a decimal (e.g. 6 thousandths = 0.006). * I can partition a decimal into its place-value parts. * I can say the value of each digit after the point.

Mini masterclass structure (25 min)

Activate (3 min): Show 2.456 on a place-value chart. Ask the value of the 6.

Model (10 min): Partition 2.456 = 2 + 0.4 + 0.05 + 0.006. Write 6 thousandths as 0.006 and 25 thousandths as 0.025.

Guided practice (8 min): Students partition decimals to 3 dp and write thousandths as decimals using a chart.

Check (4 min): Exit question: partition 3.072 and write 'eighteen thousandths' as a decimal.

Key vocabulary	Differentiation
decimal, tenths, hundredths, thousandths, partition, place value, expand	**Support:** Use a labelled place-value chart; build the number digit by digit. **Extension:** Partition decimals in more than one way and order a set of decimals to 3 dp.

Assessment	Teaching notes (annotate here)
Students express thousandths and partition decimals — links to R2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Adding and subtracting decimals

R3 | Red band | DoE Lessons 3 & 4 (adapted)

Duration: 25-30 min	**Outcomes: R3**	Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: add and subtract decimals to 3 dp

Linked to: DoE Lessons 3 & 4 (adapted) -- Lessons 3 & 4 -- Adding and subtracting decimals | Key resources: Place-value chart, decimal grids

Learning intention	Success criteria
Students are learning to: add and subtract decimals up to 3 decimal places	Students can: * I can line up the decimal points to add or subtract. * I can add a zero to fill an empty place (e.g. 3.4 as 3.40). * I can add and subtract decimals to 3 decimal places.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 2.65 + 1.4. Ask why lining up the decimal points (and place values) matters.

Model (10 min): Add 12.75 + 8.6 (write 8.60) and subtract 9.3 - 4.85 (write 9.30). Stress lining up place values.

Guided practice (10 min): Students add and subtract decimals to 3 dp, filling empty places with zeros.

Check (4 min): Exit question: 3.4 - 1.85 = ? and 2.305 + 1.7 = ?

Key vocabulary	Differentiation
decimal, decimal point, place value, line up, tenths, hundredths, thousandths	**Support:** Use grid paper and a place-value chart so digits line up. **Extension:** Add and subtract decimals with different numbers of decimal places in one problem.

Assessment	Teaching notes (annotate here)
Students add and subtract decimals to 3 dp — links to R3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Finding 10%, 25% and 50%

R1 | Red band | DoE Lessons 4 & 5 (adapted)

Duration: 25-30 min	**Outcomes: R1**	Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: 10% = ÷10, 25% = ÷4, 50% = half

Linked to: DoE Lessons 4 & 5 (adapted) -- Lessons 4 & 5 -- Benchmark percentages | Key resources: Hundred grids, mini whiteboards

Learning intention	Success criteria
Students are learning to: find 10%, 25% and 50% of a quantity	Students can: * I know 10% is one-tenth of 100%, so 10% is dividing by 10. * I can find 25% by dividing by 4 and 50% by halving. * I can find these percentages of a quantity.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a hundred grid. Shade 10%, 25% and 50%. Ask what fraction each is.

Model (10 min): 10% of 80 = 80 ÷ 10 = 8. 25% of 80 = 80 ÷ 4 = 20. 50% of 80 = half = 40.

Guided practice (10 min): Students find 10%, 25% and 50% of a range of quantities (whole numbers and money).

Check (4 min): Exit question: find 10%, 25% and 50% of 60.

Key vocabulary	Differentiation
per cent, %, 10%, 25%, 50%, divide, half, quarter, tenth, quantity	**Support:** Connect each percentage to its fraction (½, ¼, 1/10) and use a hundred grid. **Extension:** Find amounts like 75% (50% + 25%) and 5% (half of 10%).

Assessment	Teaching notes (annotate here)
Students find 10/25/50% of a quantity — links to R1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Percentages of quantities and discounts

Y4 | Yellow band | DoE Lessons 5 & 6 (adapted)

Duration: 30 min	**Outcomes: Y4**	Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: percentages as fractions/decimals; estimate discounts

Linked to: DoE Lessons 5 & 6 (adapted) -- Lessons 5 & 6 -- Percentages and discounts | Key resources: Shopping catalogues, mini whiteboards

Learning intention	Success criteria
Students are learning to: represent common percentages as fractions and decimals and estimate discounts of 10%, 25% and 50%	Students can: * I can write 10%, 25% and 50% as fractions and decimals. * I can estimate a 10%, 25% or 50% discount on a price. * I can work out the new price after a discount.

Mini masterclass structure (30 min)

Activate (4 min): A $60 jacket is '25% off'. Ask roughly how much you save.

Model (10 min): 25% = ¼ = 0.25. 25% of $60 = $15 off, so the price is $45. Estimate a 10% discount on $39 (about $4).

Guided practice (12 min): Students find discounts of 10/25/50% on catalogue prices and the new price each time.

Check (4 min): Exit question: a $40 game is 50% off. What is the discount and the new price?

Key vocabulary	Differentiation
percentage, fraction, decimal, discount, sale, estimate, new price, save	**Support:** Use the benchmark fractions and a hundred grid; round prices to estimate. **Extension:** Compare two discounts (e.g. 25% off vs $15 off) and decide which is better.

Assessment	Teaching notes (annotate here)
Students find percentage discounts — links to Y4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Estimating and checking reasonableness

Y5 | Yellow band | DoE Lesson 6 (adapted)

Duration: 25-30 min	**Outcomes: Y5**	Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: round to estimate; check reasonableness

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Estimation and reasonableness | Key resources: Mini whiteboards, number lines

Learning intention	Success criteria
Students are learning to: round numbers to estimate and use the estimate to check whether an answer is reasonable	Students can: * I can round numbers to make an estimate. * I can use an estimate to check if an answer is reasonable. * I can spot an answer that is clearly wrong.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 412 + 389 = 4801. Ask whether that looks reasonable and why not.

Model (10 min): Estimate 412 + 389 as 400 + 400 = 800. The true answer 801 is close — 4801 is unreasonable. Round to estimate first, then compute.

Guided practice (10 min): Students estimate sums and differences by rounding, then check given answers for reasonableness.

Check (4 min): Exit question: estimate 596 + 207 by rounding, then say if 803 is reasonable.

Key vocabulary	Differentiation
estimate, round, reasonable, approximate, check, nearest ten, nearest hundred	**Support:** Round each number to the nearest hundred and add the rounded numbers. **Extension:** Estimate multi-step calculations and decide how close the estimate should be.

Assessment	Teaching notes (annotate here)
Students estimate and check reasonableness — links to Y5 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Decimal word problems

Y3 | Yellow band | DoE Lessons 3, 6, 7 & 8 (adapted)

Duration: 30 min	**Outcomes: Y3**	Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: decimal word problems; justify the strategy

Linked to: DoE Lessons 3, 6, 7 & 8 (adapted) -- Lessons 3, 6-8 -- Decimal word problems | Key resources: Resource 1 - shopping receipt, mini whiteboards

Learning intention	Success criteria
Students are learning to: solve word problems involving adding and subtracting decimals and justify the strategy used	Students can: * I can pull the numbers out of a word problem and decide to add or subtract. * I can solve the problem with decimals to 3 dp. * I can explain why my strategy is appropriate.

Mini masterclass structure (30 min)

Activate (4 min): From a receipt: items cost $4.50, $2.75 and $1.20. Ask for the total and the change from $10.

Model (10 min): Total = 4.50 + 2.75 + 1.20 = 8.45. Change from $10 = 10.00 - 8.45 = 1.55. Justify lining up decimals.

Guided practice (12 min): Students solve money and measurement decimal word problems, justifying each strategy.

Check (4 min): Exit question: a bottle holds 1.25 L; 0.4 L is poured out. How much is left? Explain your method.

Key vocabulary	Differentiation
decimal, total, difference, change, justify, strategy, word problem, reasonable	**Support:** Highlight the numbers and the question; provide a place-value grid. **Extension:** Solve two-step decimal problems and write a decimal word problem for a partner.

Assessment	Teaching notes (annotate here)
Students solve and justify decimal word problems — links to Y3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Multi-step word problems

Y6 | Yellow band | DoE Lessons 7 & 8 (adapted)

Duration: 30-35 min	**Outcomes: Y6**	Syllabus: MA3-AR-01, MA3-MR-02, MAO-WM-01 | AR-B/MR-B: multistep problems, more than one operation

Linked to: DoE Lessons 7 & 8 (adapted) -- Lessons 7 & 8 -- Rich multi-step tasks | Key resources: Mini whiteboards, problem cards

Learning intention	Success criteria
Students are learning to: solve multi-step word problems that require more than one operation	Students can: * I can break a problem into steps and decide the operations. * I can solve each step and use the result in the next. * I can check my final answer is reasonable.

Mini masterclass structure (30-35 min)

Activate (4 min): You buy 3 tickets at $12 each and pay with $50. Ask how to find the change in two steps.

Model (12 min): Step 1: 3 x 12 = 36. Step 2: 50 - 36 = 14. Then a longer problem mixing +, - and a discount.

Guided practice (14 min): Students plan and solve multi-step problems, recording each step and checking with an estimate.

Check (4 min): Exit question: 4 drinks at $2.50 and 2 snacks at $3.20 — what is the total, and the change from $20?

Key vocabulary	Differentiation
multi-step, operation, step, plan, total, change, estimate, reasonable	**Support:** Provide a step-by-step planning frame; do one step at a time. **Extension:** Create a multi-step problem with at least three operations for a partner to solve.

Assessment	Teaching notes (annotate here)
Students solve multi-step word problems — links to Y6 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Any percentage of a quantity (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30 min	**Outcomes: G1**	Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: percentages of quantities in various contexts

Learning intention	Success criteria
Students are learning to: find any percentage of a quantity, not just the 10/25/50% benchmarks	Students can: * I can find 1% by dividing by 100. * I can build other percentages from 10% and 1% (e.g. 15% = 10% + 5%). * I can find any percentage of a quantity.

Mini masterclass structure (30 min)

Activate (4 min): Ask how to find 15% of $80 if you already know 10% of $80 = $8.

Model (12 min): 10% of 80 = 8, 5% = half of that = 4, so 15% = 12. Find 1% of 80 = 0.8 and build 3% = 2.4. Find 20% of 45.

Guided practice (12 min): Students find percentages like 15%, 20%, 5% and 1% of quantities by building from 10% and 1%.

Check (4 min): Exit question: find 15% of 60 and 5% of 200.

Key vocabulary	Differentiation
percentage, 1%, 10%, build, divide by 100, quantity, of	**Support:** Always find 10% first, then halve for 5% or take a tenth again for 1%. **Extension:** Find percentages like 12.5% and 35% and check using a different build.

Assessment	Teaching notes (annotate here)
Students find any percentage of a quantity — links to G1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Percentage increase and decrease (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30-35 min	**Outcomes: G2**	Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: percentage increase/decrease; GST, profit/loss

Learning intention	Success criteria
Students are learning to: calculate percentage increases and decreases, including discounts and GST	Students can: * I can decrease a price by a percentage to find a sale price. * I can increase a price by a percentage (e.g. add 10% GST). * I can solve a real-life percentage increase or decrease problem.

Mini masterclass structure (30-35 min)

Activate (4 min): A $50 item is reduced by 20%. Ask for the discount and the new price.

Model (12 min): Decrease: 20% of 50 = 10, new price = 40. Increase (GST): 10% of $30 = $3, price incl. GST = $33.

Guided practice (12 min): Students find sale prices (decrease) and prices including 10% GST (increase) for AUD amounts.

Check (5 min): Exit question: add 10% GST to a $60 part, then find a 25% discount on a $48 item.

Key vocabulary	Differentiation
increase, decrease, discount, GST, percentage, new price, original, mark-up	**Support:** Find the percentage amount first, then add or subtract it from the original. **Extension:** Apply a discount and then GST to the same amount and discuss the order.

Assessment	Teaching notes (annotate here)
Students calculate percentage increase and decrease — links to G2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Finding the whole from a percentage (Stage 4)

G3 | Green band | Original mini lesson

Duration: 30 min	Outcomes: G3	Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: unitary method, reverse percentage

Learning intention	Success criteria
Students are learning to: find the whole (original) amount when given a percentage of it, using the unitary method	Students can: * I can find 1% when I know what a percentage is worth. * I can multiply 1% by 100 to find the whole. * I can solve a reverse-percentage word problem.

Mini masterclass structure (30 min)

Activate (4 min): 20% of a number is 16. Ask how to find the whole number.

Model (12 min): Unitary method: 20% = 16, so 1% = 16 ÷ 20 = 0.8, and 100% = 0.8 x 100 = 80. Check: 20% of 80 = 16.

Guided practice (10 min): Students find the whole from a given percentage using the unitary method, then check.

Check (4 min): Exit question: 25% of a price is $12. What is the full price?

Key vocabulary	Differentiation
percentage, whole, original, unitary method, 1%, 100%, reverse, find the whole	**Support:** Always find 1% first, then multiply by 100 for the whole. **Extension:** Solve reverse-percentage problems after a discount (find the original price from a sale price).

Assessment	Teaching notes (annotate here)
Students find the whole from a percentage — links to G3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Unit 35 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Find 10%, 25% and 50% of a quantity.		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Converting Between Fractions, Decimals, and Percents | Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Worksheet (print)	Corbettmaths — Percentages of an Amount (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
R2 -- Express thousandths as decimals; partition decimals by place value.		
Video	Place Value to the Thousandths — Representing Decimal Numbers | Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Video	Expanded Form with Decimals | Math with Mr. J	https://www.youtube.com/watch?v=GX8o-S6Vxig
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R3 -- Add and subtract decimals up to 3 decimal places.		
Video	Adding and Subtracting Decimals (How to) | Math with Mr. J	https://www.youtube.com/watch?v=PnwLv6khwk8
Video	How to Add and Subtract Decimals (Step-by-Step Examples) | Math with Mr. J	https://www.youtube.com/watch?v=Cg-_TeiaSa8
Worksheet (print)	Corbettmaths — Adding Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf
Worksheet (print)	Corbettmaths — Subtracting Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
R4 -- Use start and finish times to calculate elapsed time.		
Video	Calculating Elapsed Time Using a Timeline | EasyTeaching	https://www.youtube.com/watch?v=ML6r7BEZo7M
Video	Learn How to Find Elapsed Time | Mr. Tom Teaches	https://www.youtube.com/watch?v=Ths2BdqtO4s
Worksheet (print)	Corbettmaths — Timetables (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/timetables-pdf.pdf
Y1 -- Add/subtract 3+ numbers with place value; decide when a calculator is efficient.		
Video	How to Add & Subtract Large Numbers (Without a Calculator) | Math with Mr. J	https://www.youtube.com/watch?v=186wdnoAYJc
Video	Adding Large Numbers — A Quick Review | Math with Mr. J	https://www.youtube.com/watch?v=T9asFgfN5bg
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y2 -- Apply efficient strategies (levelling, constant difference, bridging).		
Video	How to Use Bridging to Support Mental Addition and Subtraction | Third Space Learning	https://www.youtube.com/watch?v=oDDC1thyc2Q
Video	Addition and Subtraction: The Compensation Strategy | Charlotte Dooley	https://www.youtube.com/watch?v=X3kgzlb8VrM
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y3 -- Solve decimal word problems and justify the strategy.		
Video	How to Add and Subtract Decimals (Step-by-Step Examples) | Math with Mr. J	https://www.youtube.com/watch?v=Cg-_TeiaSa8
Video	Estimating Decimal Sums | Math with Mr. J	https://www.youtube.com/watch?v=x05LHZJNKQ0
Worksheet (print)	Corbettmaths — Subtracting Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
Y4 -- Represent percentages of quantities as fractions/decimals; find 10/25/50% discounts.		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Converting Between Fractions, Decimals, and Percents | Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Worksheet (print)	Corbettmaths — Percentages of an Amount (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf

Type	Resource	Link
Worksheet (print)	Corbettmaths — Fractions, Decimals, Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
Y5 -- Round to estimate and check the reasonableness of answers.		
Video	Estimating Whole Number Sums & Differences | Math with Mr. J	https://www.youtube.com/watch?v=EvQf38lnAJc
Video	Estimating Decimal Sums | Math with Mr. J	https://www.youtube.com/watch?v=x05LHZJNKQ0
Worksheet (print)	Corbettmaths — Estimation (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
Y6 -- Solve multi-step word problems requiring more than one operation.		
Video	Multi-Step Word Problem (Addition, Subtraction, Multiplication) | Khan Academy	https://www.youtube.com/watch?v=-sSDb_wZqKQ
Video	How to Add & Subtract Large Numbers | Math with Mr. J	https://www.youtube.com/watch?v=186wdnoAYJc
Worksheet (print)	Corbettmaths — Estimation (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
G1 -- Find any percentage of a quantity.		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Converting Between Fractions, Decimals, and Percents | Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Worksheet (print)	Corbettmaths — Percentages of an Amount (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
G2 -- Calculate percentage increase and decrease, incl. discounts and GST.		
Video	Increasing/Decreasing by a Percentage | Corbettmaths	https://www.youtube.com/watch?v=tUtgC7ZrsRc
Video	Percentage Change | Corbettmaths	https://www.youtube.com/watch?v=Q2gRAS08fE0
Worksheet (print)	Corbettmaths — Increasing/Decreasing by a Percentage (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Increasing-by-a-Percentage-pdf.pdf
Worksheet (print)	Corbettmaths — Percentage Change (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/percentage-change-pdf.pdf
G3 -- Find the whole from a percentage (reverse percentage, unitary method).		
Video	Percent Problems — Finding the Whole | Math with Mr. J	https://www.youtube.com/watch?v=9sMQEajxVwk
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Worksheet (print)	Corbettmaths — Reverse Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reverse-percentages-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Increasing/Decreasing by a Percentage | Corbettmaths	https://www.youtube.com/watch?v=tUtgC7ZrsRc
Video	Percent Problems — Finding the Whole | Math with Mr. J	https://www.youtube.com/watch?v=9sMQEajxVwk
Video	Percentage Change | Corbettmaths	https://www.youtube.com/watch?v=Q2gRAS08fE0
Worksheet (print)	Corbettmaths — Percentages of an Amount (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
Worksheet (print)	Corbettmaths — Increasing/Decreasing by a Percentage (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Increasing-by-a-Percentage-pdf.pdf
Worksheet (print)	Corbettmaths — Reverse Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reverse-percentages-pdf.pdf
Website	Maths is Fun — Percentages (interactive)	https://www.mathsisfun.com/percentage.html
Website	Maths is Fun — Percentage Change (interactive)	https://www.mathsisfun.com/numbers/percentage-change.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Percentages of an Amount (PDF) -- https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
- ☐ Corbettmaths — Place Value (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
- ☐ Corbettmaths — Adding Decimals (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf

- ☐ Corbettmaths — Subtracting Decimals (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
- ☐ Corbettmaths — Timetables (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/timetables-pdf.pdf
- ☐ Corbettmaths — Fractions, Decimals, Percentages (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
- ☐ Corbettmaths — Estimation (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
- ☐ Corbettmaths — Increasing/Decreasing by a Percentage (PDF) -- https://corbettmaths.com/wp-content/uploads/2018/11/Increasing-by-a-Percentage-pdf.pdf
- ☐ Corbettmaths — Percentage Change (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/percentage-change-pdf.pdf
- ☐ Corbettmaths — Reverse Percentages (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/reverse-percentages-pdf.pdf